

Homework 6 Solutions

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1. Given that $\text{proj}_{\vec{u}} \vec{v} = (3, 6, -3)$ and $\vec{u} = (1, 2, -1)$, find the possible values of $\vec{v}$.

5 points

Solution:

We use the projection formula

$$\text{proj}_{\vec{u}} \vec{v} = \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}.$$

First compute

$$\vec{u} \cdot \vec{u} = 1^2 + 2^2 + (-1)^2 = 6.$$

$$\frac{\vec{v} \cdot \vec{u}}{1^2 + 2^2 + (-1)^2} (1, 2, -1) = (3, 6, -3) \Rightarrow \frac{\vec{v} \cdot \vec{u}}{6} = 3 \Rightarrow \vec{v} \cdot \vec{u} = 18.$$

Hence, any vector $\vec{v}$ satisfying $\vec{v} \cdot \vec{u} = 18$ will produce the given projection.

If $\vec{v} = (x, y, z)$, then any vector satisfying $x + 2y - z = 18$ will produce the given projection.

2. Suppose a vector $\vec{u} = (6, 2, 1)$ that lies in $\mathbb{R}^3$. You also have a space S spanned by $\vec{v} = (1, 1, 0)$ and $\vec{w} = (1, -1, 1)$. **10+5=15**

Solution:

- Find the projection of $\vec{u}$ on S . Find the coordinates in terms of the basis vectors of S .

Solution:

Let

$$A = [\vec{v} \quad \vec{w}] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\text{proj}_S \vec{u} = A(A^T A)^{-1} A^T \vec{u} = A\vec{c}$$

$$A^T A = \begin{bmatrix} \vec{v} \cdot \vec{v} & \vec{v} \cdot \vec{w} \\ \vec{w} \cdot \vec{v} & \vec{w} \cdot \vec{w} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad (A^T A)^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}.$$

Also

$$A^T \vec{u} = \begin{bmatrix} \vec{v} \cdot \vec{u} \\ \vec{w} \cdot \vec{u} \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}.$$

The coordinates of the projection in the basis $\{\vec{v}, \vec{w}\}$ are

$$\vec{c} = (A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{5}{3} \end{bmatrix}.$$

- Find the component of $\vec{u}$ orthogonal to S .

Solution:

$$\text{proj}_S \vec{u} = A\vec{c} = 4\vec{v} + \frac{5}{3}\vec{w} = 4 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \frac{5}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{17}{3} \\ \frac{7}{3} \\ \frac{5}{3} \end{pmatrix}.$$

The orthogonal component is

$$\vec{u} - \text{proj}_S \vec{u} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{17}{3} \\ \frac{7}{3} \\ \frac{5}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

3. A robot arm in 3D moves in the directions spanned by $\vec{v} = (1, 0, 1)$ and $\vec{w} = (1, 1, 1)$. If the control force is applied as $\vec{F} = (3, 2, 5)$, find the component of $\vec{F}$ that lies within the plane of motion. Give the coordinates in terms of the basis vectors of the plane of motion. **10 points**

Solution

The basis vectors of the plane of motion are $\vec{v}$ and $\vec{w}$.

Let

$$A = [\vec{v} \quad \vec{w}] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

and

$$S = \text{Span}(\vec{v}, \vec{w})$$

Then, the component of $\vec{F}$ that lies in the plane of motion is given by

$$\text{proj}_S \vec{F} = A(A^T A)^{-1} A^T \vec{F} = A\vec{c}$$

The coordinates of the projection in the basis of the plane of motion are

$$\vec{c} = (A^T A)^{-1} A^T \vec{F} = \begin{bmatrix} 1.5 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

4. Suppose you have the following data:

x	1	2	3
y	2	2	4

Give the line of best fit of the form: $y = mx + c$. **10 points**

Solution. We fit a line $y = mx + c$ to the data $(1, 2), (2, 2), (3, 4)$ by least squares.

We have

$$\vec{y} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} m \\ c \end{bmatrix},$$

so that $\vec{y} \approx \text{proj}_S \vec{y} = A\vec{b}$ where S is spanned by the column vectors of A . The least-squares solution is

$$\vec{b} = (A^T A)^{-1} A^T \vec{y}.$$

Compute

$$A^T A = \begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}, \quad (A^T A)^{-1} = \frac{1}{6} \begin{bmatrix} 3 & -6 \\ -6 & 14 \end{bmatrix}, \quad A^T \vec{y} = \begin{bmatrix} 18 \\ 8 \end{bmatrix}.$$

Hence

$$\vec{b} = (A^T A)^{-1} A^T \vec{y} = \frac{1}{6} \begin{bmatrix} 3 & -6 \\ -6 & 14 \end{bmatrix} \begin{bmatrix} 18 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix}.$$

Therefore, the line of best fit is

$$\boxed{y = x + 0.67}.$$

5. For $x = (1, 2, 3, 4)$ and $y = (2, 4, 5, 8)$, find the best fit line through the origin. **5 points**

Solution.

We fit a line $y = mx$ to the data points $(1, 2), (2, 4), (3, 5), (4, 8)$.

Since the line must pass through the origin, we approximate

$$\vec{y} \approx m \vec{x} = \text{proj}_{\vec{x}} \vec{y},$$

so

$$m = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}}.$$

Compute

$$\vec{y} \cdot \vec{x} = 2(1) + 4(2) + 5(3) + 8(4) = 57,$$

$$\vec{x} \cdot \vec{x} = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

Thus

$$m = \frac{57}{30} = 1.9.$$

Therefore, the best fit line through the origin is

$$\boxed{y = 1.9x}.$$